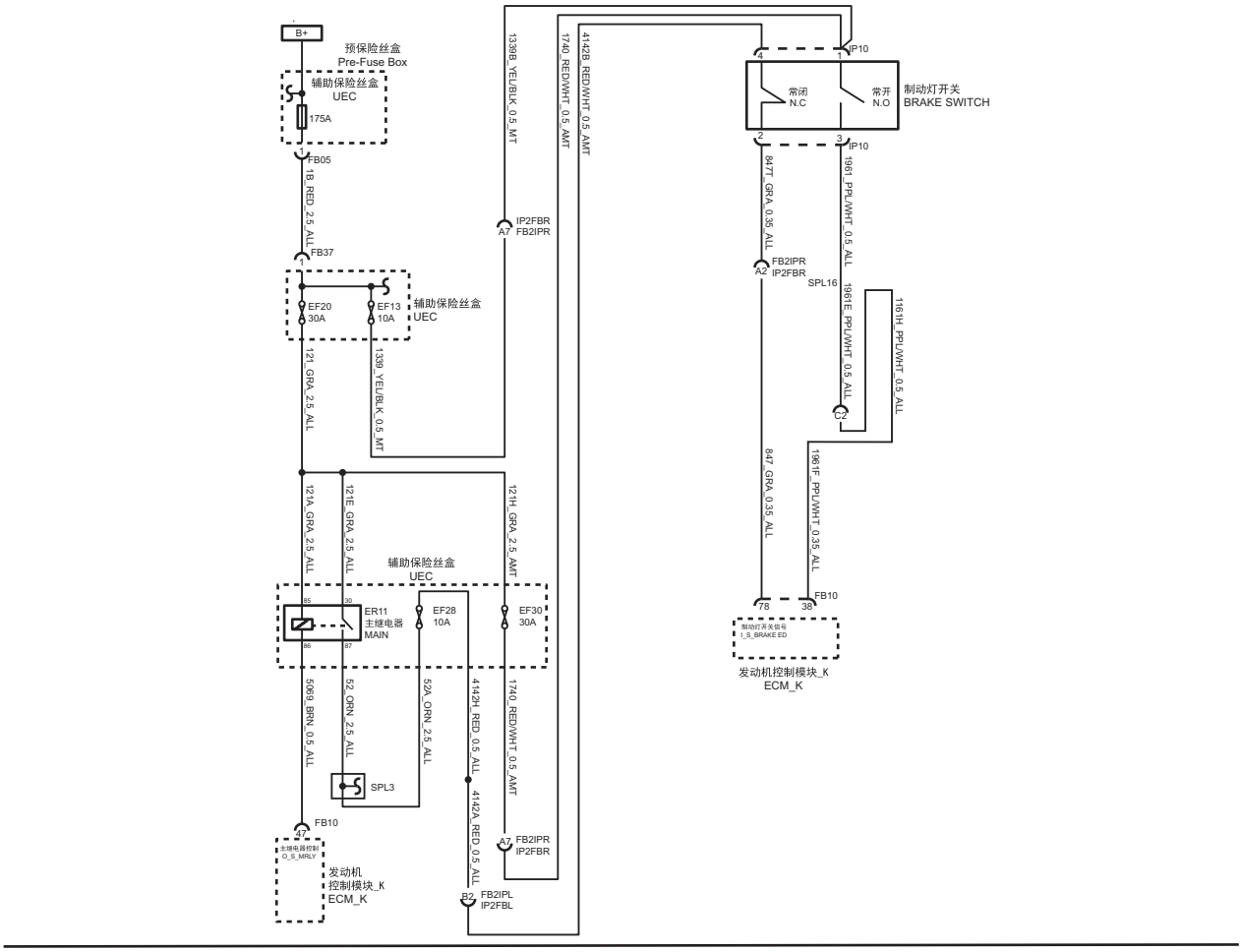
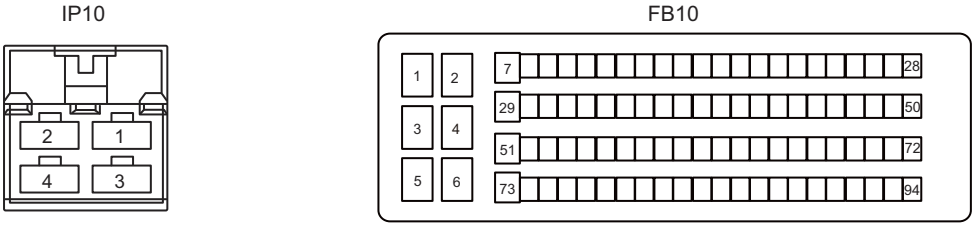
P0504

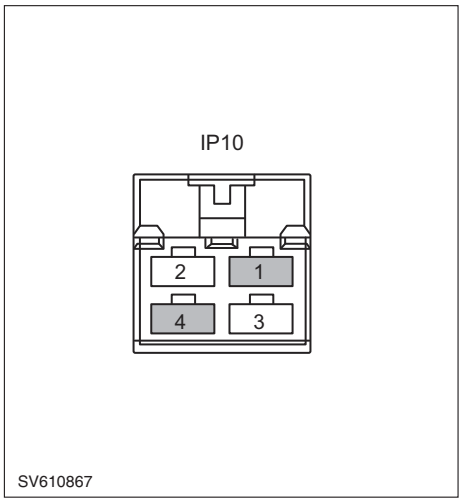
Description of fault diagnosis codes
P0504-62：Plausibility check for Brake
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Fault in harness or harness plug • Brake switch damages
Description of circuit diagram
The switch of brake lamp is installed on the bracket of brake pedal. ECM detects the status of brake pedal via such two kinds of input (brake switch ON/OFF signal).

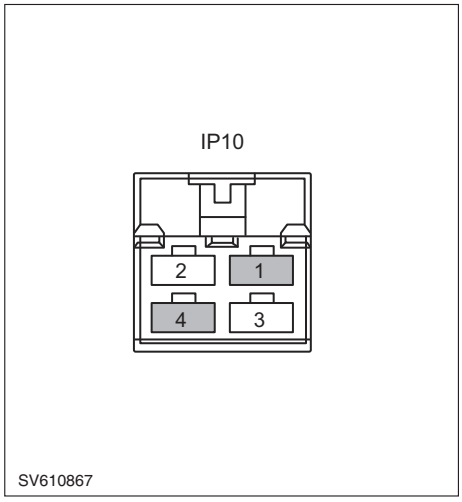
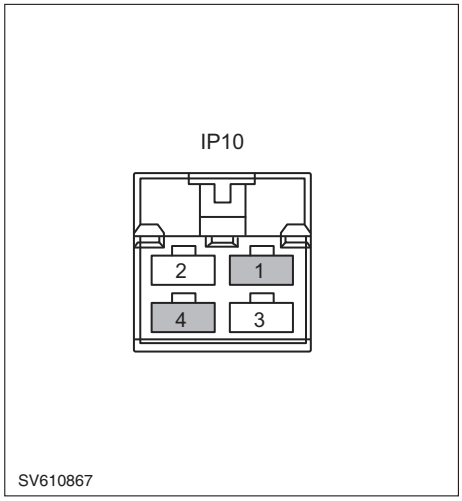
Circuit diagram

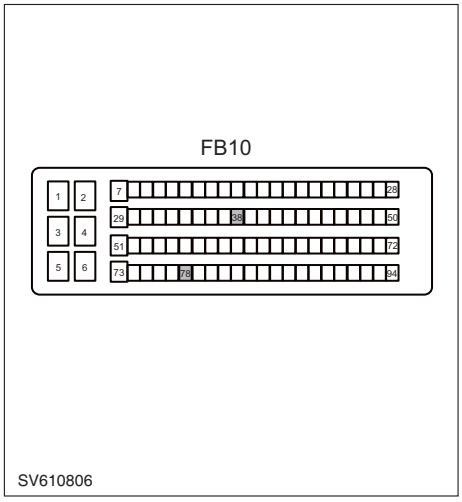
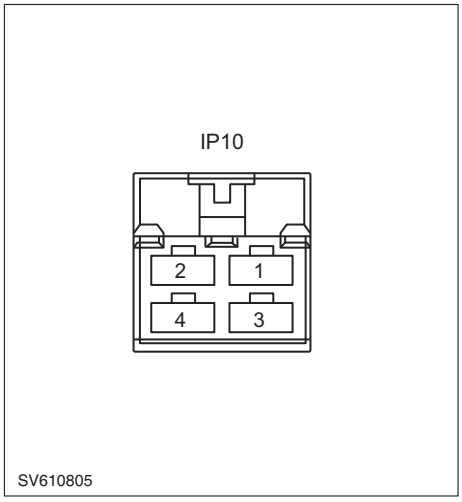




SV610804

Troubleshooting procedures	
Step 1	General inspection.
	1 Check the fuseEF13、EF20、EF28、EF30 for damage. 2 Check battery capacity. 3 Check if there is any damage, bad contact, aging and looseness phenomenon on harness plug of engine control module and brake light switch. 4 Check whether the result is normal?
	No → Repair or replace the faulted parts.
Yes ↓	
Step 2	Check fault codes.
	1 Connect the fault diagnosis instrument to diagnosis interface. 2 Turn the ignition switch to ON position. 3 Diagnose the engine control system with diagnostic apparatus. 4 If there is any fault code except for P0504?
	Yes → Reference: DTC Summary List (EU5).
No ↓	
Step 3	Check the power supply circuit of brake light switch.
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug of brake light switch. 3 Turn the ignition switch to ON position. 4 Measure the voltage between harness plug of brake light switch and ground. Measuring circuit: voltage between terminal 1 of plug IP10 and ground. Measuring circuit: voltage between terminal 4 of plug IP10 and ground. **Standard value: 10~14V** 5 Check whether the result is normal?
	Yes → Go to Step 6.
No ↓	

Step 4	Check the power supply circuit (open circuit) of brake light switch.
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug of brake light switch. 3 Unplug the main relay. 4 Measure the voltage between harness plug of brake light switch and the positive pole of main relay and storage battery. Measuring circuit: the voltage between terminal 1 of plug IP10 and the positive pole of storage battery. Measuring circuit: voltage between terminal 4 of plug IP10 and terminal 87 of main relay. **Standard value: ≈ 0Ω** 5 Check whether the result is normal?
	No → Repair or replace the faulted parts.
Yes ↓	
Step 5	Check the power supply circuit (short circuit) of brake light switch.
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug of brake light switch. 3 Unplug the main relay. 4 Measure the resistance between harness plug of brake light switch and ground. Measuring circuit: resistance between terminal 1 of plug IP10 and ground. Measuring circuit: resistance between terminal 4 of plug IP10 and ground. **Standard value: ∞** 5 Check whether the result is normal?
	No → Repair or replace the faulted parts.
Yes ↓	

Step 6	Check the brake switch circuit.
	6 Turn the ignition switch to OFF. 7 Connect the harness plug of engine control module. 8 Connect the harness plug of brake switch. 9 Turn the ignition switch to ON position. 10 Measure the voltage between harness plug terminal on engine control module and ground. Measuring circuit: the voltage between terminal 38 of plug FB10 and ground. **Standard value: ≈ 0V / Release the brake pedal** **Standard value: 10~14V / Press on the brake pedal** Measuring circuit: the voltage between terminal 78 of plug FB10 and ground. **Standard value: 10~14V / Release the brake pedal** **Standard value: ≈ 0V / Press on the brake pedal** 11 Check whether the result is normal?
	No → Repair or replace the faulted parts.
Yes ↓	
Step 7	Check the brake light switch (component inspection).
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug of brake light switch. 3 Measure the resistance between harness plug terminals of brake light switch. Measuring circuit: resistance between terminal 2 and terminal 4 of IP10 plug. **Standard value: ~0Ω/ Release the brake pedal** **Standard value: ⩾ 1MΩ / Press on the brake pedal** Measuring circuit: resistance between terminal 1 and terminal 3 of IP10 plug. **Standard value: ≈ 0Ω / Press on the brake pedal** **Standard value: ⩾ 1MΩ / Release the brake pedal** 4 Check whether the result is normal?
	No → Replace the brake light switch.
Next ↓	
Step 8	Complete the check and confirm whether the system works normally nor not.